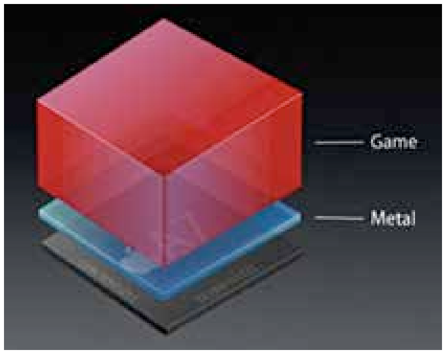
Metal is hardware

The earlier generation of hardware acceleration for graphical output on the iOS 7 systems was called OpenGL ES. In this architecture OpenGL ES was placed between the games being run on the device and Apple's A7 processing core which powered the game. The function of OpenGL ES was to translate and communicate the command functions into the graphic output commands, which were received by the hardware. This basically made it a middle-man in the system, which negotiated the messages between command and power for optimum graphical output. But it was weighed down with a lot of overhead functions that limited its effectiveness on the iOS 7 system.



Metal translates commands faster and smoother for better user experience.

With the new iOS 8 systems Apple decided to move things around and Metal was born. The name itself implies how close the "middle-man" is to the hardware. By getting in "close to the metal" or hardware Apple is able to let its devices run graphical commands faster and smoother than OpenGL ES could. As a user however knowing the intricacies of Metal isn't really that important since it only affects how developers code their apps for maximum visual quality. The system allows game developers to take the performance of iOS 8 and push it to the max to give users a stunning visual experience.

This is good news for owners of the older iOS 7 devices as well since Metal is a software enhancement and is hardware independent. Once an iPhone 5 or 5S is upgraded to iOS 8, Metal takes over from OpenGL ES to manage all graphical displays. This effect is specifically for game developers who can use both the older A7 processors as well as the new A8 processors which come with the new iPhone 6 and 6 Plus. Although casual game makers can use the new SceneKit SDK for smaller games, the big high intensity console-level graphic games are going to come alive with the help of Metal. It is however obvious that Metal will look a whole lot better on the newer 64-bit Apple A8 processors which in any case has